

COMSATS University Islamabad, Vehari Campus

Information Communication Technology (CSC101)

Lab 12

Topic: Strings (Python)

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Objective

Learn about strings in Python and practice common string operations through examples and tasks.

1. Useful Concepts

What is a String?

A string is a collection of characters enclosed in quotes (' ' or " ").

Example:

```
x = 'single line string'
y = "single line string"
```

Multiline Strings

```
y = '''This is a
multiline string'''
```

Indexing

Strings are indexed starting from 0.

```
s = "Hello"
print(s[0])
```

Output:

H

Length of String

Use `len()` function:

```
s = "Hello"
print(len(s))
```

Output:

5

Slicing

Extract part of a string:

```
s = "Python"
print(s[0:3])
```

Output:

Pyt

Concatenation

Definition: Concatenation means joining two or more strings together into one string.

```
s1 = "Hello"
s2 = "World"
result = s1 + " " + s2
print(result)
```

Output:

Hello World

Repetition

Definition: Repetition means repeating a string multiple times using the `*` operator.

```
s = "Hi"
print(s * 3)
```

Output:

HiHiHi

Membership Testing

Definition: Membership testing checks whether a specific character or substring exists in a string using `in` or `not in`.

```
s = "Welcome"
print("come" in s)
print("xyz" not in s)
```

Output:

True

True

Escape Sequences

Definition: Escape sequences are used to include special characters like quotes inside strings using backslash (`\`).

```
print("He said, \"Python is easy\"")
```

Output:

He said, "Python is easy"

2. Solved Lab Activities

Activity 1: Python program to tell the number of characters in a string

```
x = "This is Python"
print(len(x))
```

Output:

14

Activity 2: Python program to illustrate using a string with escaped characters

```
x = "\"This is Python\"", said Amanda"
print(x)
```

Output:

"This is Python", said Amanda

Activity 3: Python program to illustrate string concatenation, indexing, and slicing

```
string1 = 'Australia'
string2 = 'Pakistan won the world cup in 2012'

result = string1 + ' vs ' + string2[:8] + ' Match ' + string2[30]
print(result)
```

Output:

Australia vs Pakistan Match 2

Activity 4: Python program to illustrate membership testing

```
s1 = "Welcome"
print("come" in s1)
print("come" not in s1)
```

Output:

True

False

Activity 5: Python program to check whether a string is palindrome or not

```
s = input("Enter a string: ")

if s.lower() == s[::-1].lower():
    print("It's a Palindrome")
else:
    print("Not a Palindrome")
```

Sample Output:

```
Enter a string: MoM
It's a Palindrome
```

Activity 6: Python program to illustrate string functions

```
s = "welcome to python"
print(s.capitalize())
print(s.title())

s = "New England"
print(s.lower())
print(s.upper())
print(s.swapcase())
print(s.replace("England", "Haven"))
```

Output:

```
Welcome to python
Welcome To Python
new england
NEW ENGLAND
nEW eNGLAND
New Haven
```

3. Graded Lab Tasks

Task 1: Replace the first character of a string with 'z'

```
s = input("Enter a string: ")
```

```
if len(s) > 0:  
    s = 'z' + s[1:]
```

```
print(s)
```

Sample Output:

```
Enter a string: apple  
zpple
```

Task 2: Remove characters which have odd index values

```
s = input("Enter a string: ")
```

```
result = s[::2]  
print(result)
```

Sample Output:

```
Enter a string: Python  
Pto
```

Task 3: Swap first two characters of two strings

```
s1 = input("Enter first string: ")  
s2 = input("Enter second string: ")
```

```
new_s1 = s2[:2] + s1[2:]  
new_s2 = s1[:2] + s2[2:]
```

```
result = new_s1 + " " + new_s2  
print(result)
```

Sample Output:

Enter first string: abc
Enter second string: xyz
xyc abz

Task 4: Remove 0th, 4th and 5th elements from a list

```
lst = [10, 20, 30, 40, 50, 60, 70]
```

```
result = [lst[i] for i in range(len(lst)) if i not in (0, 4, 5)]  
print(result)
```

Output:

```
[20, 30, 40, 70]
```

End of Lab Notes